

Problem Set 3, Part I

Please start your answer to each problem on a new page, as we have done below!

Problem 1: Our Rectangle class revisited

1-1)

type of method: mutator

Header: public void scale(int ratio)

1-2)

type of method: accessor

Header: public bool smallerThan(Rectangle toCompare)

1-3)

problems in code: r1.height is a private variable in the Rectangle class, so it is not accessible or mutable from another class.

rewritten version:

Java

```
Rectangle r1 = new Rectangle(60, 80);  
System.out.println("r1's width is: " + r1.width);  
r1.setHeight(r1.getHeight+20);  
System.out.println(r1);    // print the new dimensions
```

Problem 2: A class that needs your help

2-1) *Revise the code found below:*

Java

```
public class Pair {  
    private int x;  
    private double y;  
  
    public double difference() {  
        return this.x - this.y;  
    }  
    /*
```

the method can't access the fields this.x and this.y because we are trying to use "this" in a static method.

"this" refers to the current instance of a class, while "static" methods refer to the class itself, not an instance.

To fix this problem, I needed to remove the "static" from the method header and turn it into an instance method

```
*/  
// add the new methods here  
public int getX(){  
    return x;  
}  
public double getY(){  
    return y;  
}  
public void setX(int newX){  
    x = newX;  
}  
public void setY(double newY){  
    y = newY;  
}
```

```
public Pair(int newX, double newY){  
    x = newX;  
    y = newY;  
}  
}
```

2-2)

Problem 3: Static vs. non-static

3-1)

type and name of the variable	static or non-static?	purpose of the variable, and why it needs to be static or non-static
double rawScore	non-static	stores the raw score associated with a given Grade object; needs to be non-static so every Grade object will have its own instance of this variable
String gradeType	non-static	stores whether this Grade object is an "assignment", "quiz", or "exam"; must be non-static so each Grade object can have its own category
static int assignmentCount	static	keeps track of the total number of Grade objects created with category "assignment"; static so the count is shared across all objects
static int quizCount	static	keeps track of the total number of Grade objects created with category "quiz"; static so the count is shared across all objects
static int examCount	static	keeps track of the total number of Grade objects created with category "exam"; static so the count is shared across all objects

3-2)

a) static or non-static?: non-static

explanation: every grade has it's own type. It has to be able to access the gradeType. A static method wouldn't be able to know which variable to change as it cannot use the "this" keyword.

b) changes it would need to make:

1. The object is no longer a quiz, therefore quizCount has to be decreased by 1
2. Change the grade's gradeType to "exam"
3. Increase the counter for examCount

3-3)

a) static or non-static?: static

Explanation: this method only uses its parameters to do the calculation

b) example of calling it: `gradeObject.computePercent(30.0, 50.0);`

3-4)

a) static or non-static?: non-static

explanation: it modifies `rawScore`, an instance variable.

b) example of calling it: `gradeObject.addExtraCredit(5.0);`